Quantification: 872_wt_hp_z03
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	8
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 8)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1305.5	1243.5	612.612	px^2
major_axis_length_px	lacuna shape	66.2315	68.6165	14.1596	px
minor_axis_length_px	lacuna shape	25.7426	25.8032	9.8277	px
aspect_ratio	lacuna shape	2.8467	2.4831	1.1271	unitless
eccentricity	lacuna shape	0.9082	0.9153	0.0591	unitless
solidity	lacuna shape	0.8777	0.8936	0.0449	unitless
roots_count	headline	9.625	8.5	3.8522	count
ring_length_r30_px	headline	404.375	408	105.261	px
ring_length_r60_px	reported	974.25	954	268.193	px
owned_length_px	ownership-dependent	2156.52	1926.11	1510.63	px
edge_count	ownership-dependent	130.25	115	98.9152	count
mean_edge_length_px	ownership-dependent	17.5432	16.9671	1.6367	px
perimeter_px	normalised	182.492	186.409	44.5499	px
ring_area_r30_px2	normalised	7554.75	7821.5	1025.95	px^2
ring_area_r60_px2	normalised	20433.4	20698.5	1840.77	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0535526	0.0500437	0.0123585	px^-1
ring_density_r60	normalised	0.0475627	0.0443069	0.0118199	px^-1
roots_per_100px_perimeter	normalised	5.4146	4.9607	1.8856	per 100 px
ring_attached_length_r30_px	network v2	362.875	353.5	117.767	px
ring_attached_length_r60_px	network v2	756	692.5	330.251	px
ring_length_w_r30_px	network v2	464.331	463.712	121.169	px
ring_length_w_r60_px	network v2	1111.78	1068.02	307.508	px
sholl_crossings_r10	network v2	7.75	7.5	2.0529	count
sholl_crossings_r20	network v2	10.125	10	2.997	count
sholl_crossings_r30	network v2	13.375	13.5	3.5832	count
loop_count	loops, report only	12.75	7	13.1339	count
loop_edge_share	loops, report only	0.2845	0.268	0.1665	share
ring_width_mean_r30_px	width	6.8037	6.3359	1.009	px

Per field
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measure	value	units
analysed_area_px2	917032	px^2
total_skeleton_length_px	26202	px
canalicular_length_density_per_px	0.0285726	px^-1
junction_count	723	count
junction_density_per_px2	0.00078841	px^-2
lacuna_count	11	count

lacunae_per_px2	1.2e-05	px^-2
skeleton_component_count	308	count
mean_component_length_px	85.0714	px
median_component_length_px	17	px
field_length_density_w_per_px	0.0322692	px^-1
field_density_without_flagged_per_px	0.0253018	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	331	px
unattached_component_count	22	count
canalicular_length_density_attached_per_px	0.0285726	px^-1
field_length_density_with_unattached_per_px	0.0289336	px^-1
excluded_area_px2	115300	px^2
analysed_field_area_px2	933276	px^2
loop_count	136	count
loop_edge_share	0.2682	share
edge_count	1913	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.1919	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	26021	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	794	69.6134	15.6156	4.4579	0.9745	0.834	-1.442	210.909	115.354	False
2	420	36.8904	15.1005	2.443	0.9124	0.8955	-0.9718	230.283	755.805	False
3	1078	83.4971	17.8088	4.6885	0.977	0.8021	-1.3939	282.36	422.258	False
4	2506	64.9879	57.5952	1.1284	0.4632	0.8735	0.3596	332.631	998.485	True
5	2428	74.728	42.3101	1.7662	0.8243	0.9324	-1.5637	398.271	285.478	False
6	2726	99.126	35.5862	2.7855	0.9333	0.9172	-1.4054	396.683	51.522	True
7	1386	67.6197	26.7997	2.5232	0.9181	0.9053	-1.5154	396.681	828.755	False
8	1506	75.3481	26.717	2.8202	0.935	0.8917	-1.4312	511.238	191.668	False
9	1101	57.7118	24.8895	2.3187	0.9022	0.9137	-1.3389	569.522	707.45	False
10	568	28.9763	25.6237	1.1308	0.4669	0.9578	-0.8322	590.484	11.2324	True
11	1731	64.4432	36.6995	1.756	0.822	0.8469	-1.4227	651.578	232.505	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	269	574	660.115	33	20.0035	6.3223
2	7	264	719	2572.84	156	16.4926	5.8721
3	8	427	908	749.156	39	19.2091	5.7143
4	7	361	807	2061.74	115	17.9282	6.0571
5	11	389	1000	1279.37	74	17.2888	7.9221
6	5	271	466	491.09	28	17.5389	5.8954

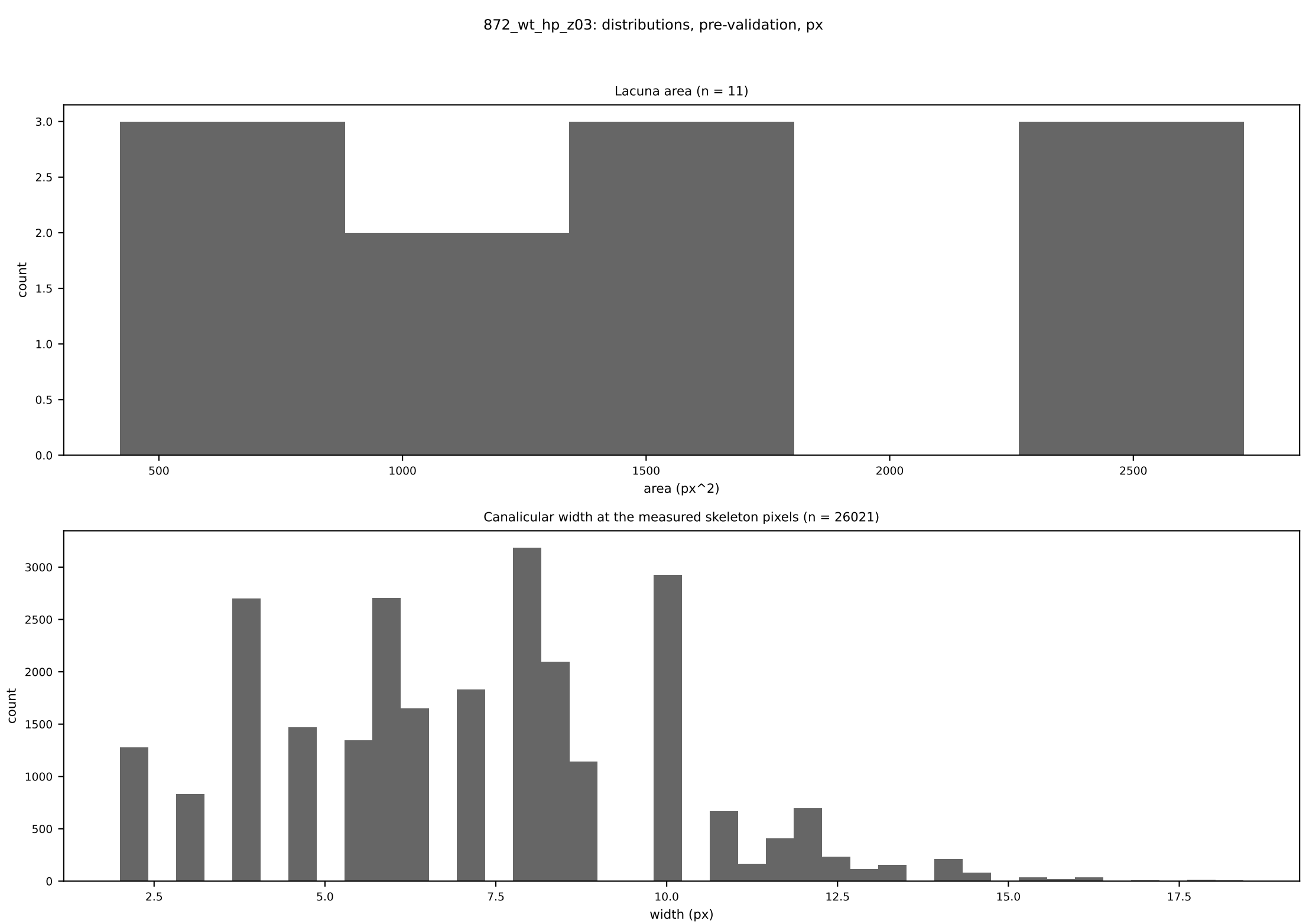
7	9	441	1195	2613.32	157	16.6454	6.2283
8	7	375	861	854.98	45	18.9996	6.3495
9	11	500	1130	4103.79	255	16.0933	8.2753
10	3	155	482	1219.13	63	19.3513	8.5766
11	18	570	1407	4418.57	283	15.6133	7.7454

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	165.782	7287	20197	1	1	0.036915	0.0284201	3.6192
2	97.0538	5433	16508	1	1	0.0485919	0.0435546	7.2125
3	229.8	8292	22152	1	1	0.0514954	0.0409895	3.4813
4	283.664	6163	14859	0.6902	0.6384	0.0585754	0.0543105	2.4677
5	223.179	8740	22193	1	1	0.044508	0.0450593	4.9288
6	253.279	8248	19209	0.8394	0.7603	0.0328565	0.0242595	1.9741
7	180.267	7666	20955	1	1	0.0575267	0.057027	4.9926
8	192.551	7977	20442	1	1	0.0470102	0.0421192	3.6354
9	151.782	6950	19530	1	1	0.0719425	0.0578597	7.2472
10	92.7696	3398	9619	0.6214	0.5799	0.0456151	0.0501092	3.2338
11	219.522	8093	21490	1	1	0.0704312	0.0654723	8.1996

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	222	335	297.409	641.044	6	4	9
2	227	453	316.676	849.306	6	10	8
3	323	555	484.647	1010.74	7	10	13
4	346	714	415.161	926.879	10	10	8
5	367	667	442.777	1125.31	6	13	13
6	218	332	297.037	514.304	5	7	9
7	345	933	509.688	1359.33	8	10	14
8	362	718	430.747	1002.25	8	9	15
9	497	1070	578.76	1287.76	9	14	16
10	155	438	180.439	545.889	4	4	6
11	560	1317	653.943	1618.54	12	11	19


872_wt_hp_z03: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 26021)
count
width (px)